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Introduction

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1.1 | Symmetry Groups in Quantum Physics

Symmetry transformations are operations carried out on the devices which prepare the states and the apparatuses which measure the observables of a quantum system having the property of not changing the results of experiments. For instance, space translations are symmetry transformations since the uniform translation of all preparations devices and measuring apparatuses does not change the results of experiments due to the homogeneity of space. Likewise, space rotations are symmetry transformations since the uniform rotation of all preparations devices and measuring apparatuses does not change the results of experiments due to the isotropy of space. In relativity, Lorentz boosts are symmetry transformations since as the result of experiments are the same in a given inertial frame and any inertial frame moving at uniform speed with respect to the first because the isotropy of spacetime.

In general, symmetry transformations gather together in families with special properties. Let G be one such family. Carrying out successively two symmetry transformations g, g' of G yields a third symmetry transformation $g' \cdot g$ of G called the **product** of g, g'. The effect of a symmetry transformation g of G can be undone by another symmetry transformation g^{-1} of G called the **inverse** of g. There exists a distinguished symmetry transformation e of G that has no effect called **neutral**. The multiplication and inversion operations and neutral element of G have the following properties:

- **Associativity**: for any three symmetry transformations g, g', g'' of G we have
$$(g'' \cdot g') \cdot g = g'' \cdot (g' \cdot g). \tag{1.1.1}$$
- **Neutrality**: there exists a symmetry transformation e of G such that for any symmetry transformation g of G we have
$$e \cdot g = g \cdot e = g. \tag{1.1.2}$$
- **Invertibility**: for any symmetry transformation g of G there exists a symmetry transformation g^{-1} of G such that
$$g^{-1} \cdot g = g \cdot g^{-1} = e. \tag{1.1.3}$$

These relations characterize G as an algebraic structure called a **group**. Since G is constituted by symmetry transformations, G is called more specifically a **symmetry group**. A group G is called **finite** if it is constituted by a finite number of elements, else it is called **infinite**. A group G is called **Abelian** if multiplication turns out to be commutative, that is
$$g \cdot g' = g' \cdot g, \quad \forall g, g' \in G. \tag{1.1.4}$$

Else, it is called non Abelian. The groups encountered in quantum physics are bot finite and infinite and Abelian and non Abelian.

TODO: Complete the example

Example (Space Translation Symmetry). [...]

TODO: Complete the example

Example (Space Rotation Symmetry). [...]

1.1.1 | Symmetry Groups and Their Operator Realization

In quantum theory, states and observables of a system are represented respectively by normalized vectors defined up to a multiplicative phase in a Hilbert space $\mathcal{H}$ and selfadjoint operators in the operator algebra $L(\mathcal{H})$. The invariance of experimental results under the action of a symmetry group reduces to the invariance of expectation values: if Ψ is a state, O is an observable and g is a symmetry transformation of G, ${}^{g}\Psi$ is the transformed state of Ψ and ${}^{g}O$ is the transformed observable of O, one must have

$$\langle {}^{g}\Psi | {}^{g} O \, | {}^{g}\Psi \rangle = \langle \Psi | \, O \, | \Psi \rangle \, . \tag{1.1.5}$$

Let us now analyze the implications of the invariance of expectation values under the action of a symmetry group G with a simple example.

Example (“yes–no” observables). As is well–known, there is a class of elementary observables, the “yes–no” observables, defined by taking the value 1, 0 according to whether the system is in a certain state Φ or not. The associated operator is given by

$$O_{\Phi}\Psi = \Phi \, \langle \Phi | \Psi \rangle \, .$$

It is natural to assume that under the symmetry transformation g of G the observable O_{Φ} transforms accordingly with the state Φ as follows:

$${}^{g}O_{\Phi} = O_{{}^{g}\Phi}.$$

The expectation value of O_{Φ} in the state Ψ' is given by the transition probability from Ψ' to Φ:

$$\langle \Psi' | \, O_{\Phi} \, | \Psi' \rangle = | \, \langle \Phi | \Psi' \rangle \, |^{2},$$

such that the **invariance of expectation values** under the action of the symmetry group G implies that

$$| \, \langle {}^{g}\Phi | {}^{g}\Psi' \rangle \, |^{2} = | \, \langle \Phi | \Psi' \rangle \, |^{2}.$$

This means that the symmetry transformation g of G must be implemented by a unitary or antiunitary operator U_g on the Hilbert space $\mathcal{H}$ of the system such that

$$| {}^{g}\Psi \rangle = U_g \, | \Psi \rangle \, , \quad {}^{g}O = U_g O U_g^{-1}.$$

In this example we have highlighted the importance of the **Wigner's theorem**:

Theorem 1.1 (Wigner's theorem)**.** *Let $\mathcal{H}$ be a Hilbert space and let G be a symmetry group of a quantum system represented on $\mathcal{H}$. Then, for each symmetry transformation g of G, there exists a unitary or antiunitary operator U_g on $\mathcal{H}$ such that*

$$\begin{cases} {}^{g}\Psi = U(g)\Psi, \\ {}^{g}O = U_g O U_g^{\dagger}, \end{cases} \tag{1.1.6}$$

since U_g is unitary or antiunitary, $U_g^{-1} = U_g^\dagger$, and so we can write the transformation of observables as $O \mapsto U_g O U_g^\dagger$. This implies that the expectation value of any observable O in any state Ψ is invariant under the action of the symmetry group G.

This is a fundamental result in quantum theory since it allows us to represent symmetry transformations as unitary or antiunitary operators on the Hilbert space of the system.

Recall that a unitary operator U on a Hilbert space $\mathcal{H}$ is a linear operator that preserves the Hilbert inner product,

$$U(\Phi\, a,\, \Phi'\, a') = U(\Phi)a,\, U(\Phi')a'$$
$$\langle U\Phi | U\Phi' \rangle = \langle \Phi | \Phi' \rangle \,,$$

while an antiunitary operator A on a Hilbert space $\mathcal{H}$ is an antilinear operator that preserves the Hilbert inner product,

$$A(\Phi\, a,\, \Phi'\, a') = A(\Phi)a^*,\, A(\Phi')a'^*$$
$$\langle A\Phi | A\Phi' \rangle = \langle \Phi | \Phi' \rangle^* \,.$$

in both cases the adjoint operator $U^\dagger$ (or $A^\dagger$) of a unitary (or antiunitary) operator U (or A) are equal to the inverse operator U^{-1} (or A^{-1}):

$$\begin{cases} \langle \Psi' | U\Psi \rangle = \left\langle U^\dagger \Psi' \middle| \Psi \right\rangle, \\ \langle \Psi' | A\Psi \rangle = (\left\langle A^\dagger \Psi' \middle| \Psi \right\rangle)^*, \end{cases} \qquad \Longrightarrow \qquad \begin{cases} U^{-1} = U^\dagger, \\ A^{-1} = A^\dagger, \end{cases}$$

respectively for the linear and antilinear case. In what follows, we leave aside the antiunitary case, which occurs only for symmetry transformations involving time reversal, and concentrate on the unitary one. Letting the symmetry transformation g vary in the group G, we obtain a family of unitary operators $U(g)$ contained in the operator algebra $L(\mathcal{H})$.

Let us come back to relation $(1.1.6)_1$. We recall again that state vectors are normalized vectors defined up to a multiplicative phase factor. The content of $(1.1.6)_1$ is that phase choice can be adjusted so that for any symmetry transformation g the state correspondence $\Psi \mapsto^g \Psi$ is codified by a unitary operator $U(g)$. This however does not completely fix $U(g)$: we are still free to replace $U(g)$ by

$$\hat{U}(g) = \alpha(g) U(g), \tag{1.1.7}$$

where $\alpha(g)$ is a complex number of unit modulus, which is called a **phase factor**. This involves a phase redefinition

$$^g\Psi \to \alpha(g)\, {}^g\Psi,$$

in $(1.1.6)_1$, which does not change the physical content of the transformation since states are defined up to a multiplicative phase factor. By a suitable choice of the phase factors $\alpha(g)$ many relations simplyfy.

It is natural and possible to require

$$^e\Psi = \Psi, \quad \forall \Psi \in \mathcal{H},$$

so that the neutral element e of the symmetry group G is represented by the identity operator

$$U(e) = \mathbb{I},$$

on the Hilbert space $\mathcal{H}$ of the system. With this choice, we can now analyze the implications of the group properties of G on the family of unitary operators $U(g)$. Considering two symmetry transformations g, g' of G, and a physical state Ψ, we have ${}^{g'}({}^{g}\Psi)$ and ${}^{g'\cdot g}\Psi$ defined only up to a conventional choice of a multiplicative **phase factor** as seen above, one cannot expect that a relation of the form ${}^{g'}({}^{g}\Psi) = {}^{g'\cdot g}\Psi$ to hold strictly in general, but only up to some phase factor

$$\gamma(g, g', \Psi) \in \mathbb{C}, \quad |\gamma(g, g', \Psi)| = 1,$$

depending on the symmetry transformations g, g' and the state Ψ. Thus we can write

$${}^{g'}({}^{g}\Psi) = {}^{g'\cdot g}\Psi\, \gamma(g, g', \Psi).$$

Let us then express an important result concerning the phase factor $\gamma(g, g', \Psi)$ appearing in the above relation.

Proposition 1.2. *The phase factor $\gamma(g, g', \Psi)$ can be chosen to be independent of the state Ψ, thus obtaining a relation of the form*

$$U(g \cdot g') = \gamma(g, g')U(g)U(g'), \tag{1.1.8}$$

which is called a ***projective representation*** *of the symmetry group G. If the phase factor $\gamma(g, g')$ can be chosen to be 1 for all g, g' in G, then we have a* ***unitary representation*** *of the symmetry group G.*

TODO: Complete the proof if needed

Proof. Proof... □

The phase factors $\gamma(g, g')$ appearing in the projective representation of the symmetry group G are not arbitrary, but must satisfy some consistency conditions, for example the *2-cocyclicity*:

Proposition 1.3. *Given three symmetry transformations g, g', g'' of G, the following relation must hold:*

$$\gamma(g, g' \cdot g'')\gamma(g', g'') = \gamma(g \cdot g', g'')\gamma(g, g'), \tag{1.1.9}$$

where functions $\gamma : G \times G \to U(1)$ satisfying the 2-cocyclicity condition are called ***2-cocycles*** *of G.*

The 2-cocycle condition is a consequence of the associativity property of the group multiplication of G.

TODO: Complete the proof if needed

Proof. Proof... □

Proposition 1.4. *The G 2-cocycle $\gamma(g, g')$ enjoys a few special properties, which albeit technical are often useful at various points:*

$$\begin{aligned} \gamma(e, g) = \gamma(g, e) = 1, \\ \gamma(g, g^{-1}) = \gamma(g^{-1}, g), \end{aligned} \tag{1.1.10}$$

for any symmetry transformation g of G.

The first property is a consequence of the neutrality property of the group multiplication of G, while the second one is a consequence of the invertibility property of the group multiplication of G.

TODO: Complete the proof if needed

Proof. Proof... □

Relation (1.1.9) entails also that, for any symmetry transformation g, the operators $U(g^{-1})$ and $U(g)^{-1}$ differ by a g-dependent phase factor. Specifically, the relation between them takes the form

$$U(g^{-1}) = \varpi(g)U(g)^{-1}, \quad \varpi(g) = \gamma(g, g^{-1})^{-1}.$$

Proof. Proof... $\square$ TODO: Complete the proof if needed

In general it is not possible to absorb the phase $\gamma(g', g)$ by redefining the operators $U(g)$ into the operators $\hat{U}(g)$ given in (1.1.7) with $\alpha(g)$ a suitable phase function. By doing so $\gamma(g', g)$ gets replaced by

$$\hat{\gamma}(g, g') = \gamma(g, g')\alpha(g \cdot g')\alpha(g)^{-1}\alpha(g')^{-1}, \tag{1.1.11}$$

and in general this cannot be made equal to 1 by a judicious choice of $\alpha(g)$. A phase function $\gamma(g', g)$ of the form

$$\gamma(g, g') = \alpha(g \cdot g')\alpha(g)^{-1}\alpha(g')^{-1}, \tag{1.1.12}$$

for some function $\alpha : G \to U(1)$ is called a **2-coboundary** of G.

Proposition 1.5. *A G 2-coboundary is a G 2-cocycle, that is, any function $\gamma(g, g')$ of the form* (1.1.12) *satisfies the 2-cocyclicity condition* (1.1.9)*.*

Proof. Proof... $\square$ TODO: Complete the proof if needed

The G 2–cocycle $\hat{\gamma}(g', g)$ in (1.1.11) can be rendered equal to 1 precisely when the G 2–cocycle $\gamma(g', g)$ is a G 2–coboundary of the form (1.1.12). This may or may not happen. The study of these matters is the object of a branch of algebra known as group cohomology. Some special results will be obtained below. At any rate, if $\hat{\gamma}(g', g)$ can be made equal to 1, the representation $\hat{U}(g)$ becomes genuinely unitary, so that

$$\hat{U}(g \cdot g') = \hat{U}(g)\hat{U}(g'),$$

for all symmetry transformations g, g' of G. Then the unitary family $U(g)$ forms a genuine **unitary representation** of the symmetry group G.

1.2 | Dynamical Symmetry Groups

Suppose that for a symmetry group G, the following is observed: by acting with any symmetry transformations of G on the devices that prepare the states of the system but not the apparatus that measures energy, the results of energy measurements do not change. The symmetry is then called **dynamical**, because *energy is the infinitesimal generator of time shifts*.

We note that also by acting with any symmetry transformations of G on both the devices preparing the states of the system and the apparatus measuring energy, the results of energy measurements do not change. Therefore, when G is a dynamical symmetry group, then the system's Hamiltonian H satisfies for any symmetry element g of G the relation

$$^{g}H = H. \tag{1.2.1}$$

If G acts on th Hilbert space through a unitary representation $U(g)$, then the invariance of the Hamiltonian under the action of G can be written as

$$U(g)HU(g)^{\dagger} = H, \quad \forall g \in G, \tag{1.2.2}$$

by virtue of Wigner's theorem $(1.1.6)_2$. This means that the Hamiltonian H commutes with all the unitary operators $U(g)$ representing the symmetry transformations of G:

$$[H, U(g)] = 0, \quad \forall g \in G.$$

This is a non trivial property severely constraining the form of H and thus providing useful structural information about the form of the interactions the system is involved in. Let us illustrate this point with a few examples taken from quantum theory.

Example (The Hydrogen Atom). The hydrogen atom is a system constituted by a proton and an electron interacting through the Coulomb potential. The Hamiltonian of the hydrogen atom is given by

$$H = \frac{\mathbf{p}^2}{2m}V(\mathbf{q}).$$

Direct observation shows that all proper rotations of the devices preparing the states of the atom leave the results of all energy measurements unchanged. Therefore, the proper rotation group SO(3) is dynamical. The atom has spherical symmetry, so from (1.2.2) we have that the Hamiltonian ha to remain unchanged under the action of any proper rotation $\mathbf{R}$ of SO(3):

$$U(\mathbf{R})HU(\mathbf{R})^{\dagger} = H, \quad \forall \mathbf{R} \in \mathrm{SO}(3).$$

It is sensible to assum that $^{\mathbf{R}}\mathbf{q} = \mathbf{R}^{-1}\mathbf{q}$ and $^{\mathbf{R}}\mathbf{p} = \mathbf{R}^{-1}\mathbf{p}$ under the action of a proper rotation. Then, we can write

$$U(\mathbf{R})\mathbf{q}U(\mathbf{R})^{\dagger} = \mathbf{R}^{-1}\mathbf{q}, \quad U(\mathbf{R})\mathbf{p}U(\mathbf{R})^{\dagger} = \mathbf{R}^{-1}\mathbf{p},$$

which means that the position and momentum operators transform as vectors under the action of SO(3).[1]

Using this last relation along with the invariance of the Hamiltonian under the action of SO(3) implies that

$$\frac{(\mathbf{R}^{-1}\mathbf{p})^2}{2m} + V(\mathbf{R}^{-1}\mathbf{q}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{q}), \quad \forall \mathbf{R} \in \mathrm{SO}(3).$$

[1] This is a consequence of the fact that they transform according to the fundamental representation of SO(3): $\mathbf{q} \mapsto \mathbf{R}^{-1}\mathbf{q}$, $\mathbf{p} \mapsto \mathbf{R}^{-1}\mathbf{p}$.

In particular, knowing that the rotations preserve the length of vectors, we have $(\mathbf{R}^{-1}\mathbf{p})^2 = \mathbf{p}^2$, so that the invariance of the Hamiltonian under the action of SO(3) reduces to

$$V(\mathbf{R}^{-1}\mathbf{q}) = V(\mathbf{q}) \qquad \Longrightarrow \qquad V(\mathbf{q}) = v(|\mathbf{q}|),$$

so that the potential V must be a function of the length of the position vector $v(|\mathbf{q}|)$ only. This is a non trivial constraint on the form of the potential V and thus on the form of the interactions the system is involved in; it is not sufficient to yield the Coulombic expression of the potential, but it is a necessary condition for it. The spherical symmetry of the hydrogen atom is a consequence of the dynamical symmetry of SO(3).

Example (A System of Identical Particles).

TODO: Complete the example

Example (Conduction Electrons in a Crystal Lattice).

TODO: Complete the example

1.2.1 | Energy Eigenvalues Classification

The energy spectrum of a quantum system typically contains many energy values. The problem then arises of devising a classification method for the energy eigenvalues. There does not exist any a priori ideal classification scheme. However, symmetry provides a natural one. In this section, we show that when the system is characterized by a dynamical symmetry group G, the energy eigenvalues can be efficiently classified based on the unitary representations of G according to which the energy eigenstates transform under the action of G.

We assume that the system's Hamiltonian H has a **completely discrete spectrum** and that each energy eigenvalue has **finite degeneracy**. Let E be an energy eigenvalue of H and let $\mathcal{H}_E$ be the corresponding eigenspace, which is d_E-dimensional given a degeneracy d_E; this subspace is spanned by all vectors Ψ which respect the eigenvalue equation

$$H\Psi = E\Psi.$$

Let now Ψ be a vector of $\mathcal{H}_E$, then for any symmetry transformation g of G we have

$$HU(g)\Psi = U(g)H\Psi = EU(g)\Psi,$$

as a direct consequence of (1.2.2). This means that $U(g)\Psi$ is also an eigenvector of H with the same eigenvalue E. Thus, the eigenspace $\mathcal{H}_E$ is invariant under the action of G. The unitary operators $U(g)$ representing the symmetry transformations of G on the Hilbert space $\mathcal{H}$ of the system restrict to unitary operators on the eigenspace $\mathcal{H}_E$, thus providing a unitary representation of G on $\mathcal{H}_E$. The energy eigenvalue E can be classified according to the unitary representation of G according to which the energy eigenstates in $\mathcal{H}_E$ transform under the action of G.

Unitary Representations on the Eigenspaces

If we now consider $\{\Phi_i\}$ an ON basis for the subspace $\mathcal{H}_E$, then these vectors respect the eigenvalue equation and the orthonormality condition

$$\begin{cases} H\Phi_i = E\Phi_i, \\ \langle \Phi_i | \Phi_j \rangle = \delta_{ij}. \end{cases}$$

Since the unitary representation of G on $\mathcal{H}_E$ is given by the restriction of the unitary representation of G on $\mathcal{H}$, we have that for any symmetry transformation g of G and any vector Φ_i of the ON basis $\{\Phi_i\}$ of $\mathcal{H}_E$, the transformed vector $U(g)\Phi_i$ is a linear combination of the vectors of the ON basis $\{\Phi_i\}$:

$$U(g)\Phi_i = \sum_{j=1}^{d_E} \Phi_j D_{ji}(g), \tag{1.2.3}$$

where $D_{ji}(g)$ are the complex, g-dependent matrix elements of the unitary representation of G on $\mathcal{H}_E$. We aim to prove that the family of matrices $D(g)$ with elements $D_{ji}(g)$ forms a unitary representation of the symmetry group G on the eigenspace $\mathcal{H}_E$.

Proposition 1.6. *For each element g of the symmetry group G, there exists a $d_E \times d_E$ complex matrix $D(g) = (D_{ji}(g))$ which is unitary:*

$$\begin{aligned} D(g \cdot g') &= D(g)D(g'), \\ D(g)^\dagger D(g) &= D(g)D(g)^\dagger = \mathbb{I}, \end{aligned} \tag{1.2.4}$$

for all g, g' in G. The family of unitary matrices $D(g)$ forms a unitary representation of the symmetry group G on the eigenspace $\mathcal{H}_E$ corresponding to the energy eigenvalue E.

Proof. As $U(g)$ is a unitary representation of G on the Hilbert space $\mathcal{H}$ of the system, we know that for any g, g' in G we have

$$U(g \cdot g')\Phi_i = U(g)U(g')\Phi_i.$$

Since the vectors Φ_i form a basis for the eigenspace $\mathcal{H}_E$, the above equation implies that the matrix elements of $U(g)$ in this basis satisfy the same multiplication rule as the matrices $D(g)$, from (1.2.3):

$$U(g \cdot g')\Phi_i = \sum_{j=1}^{d_E} \Phi_j D_{ji}(g \cdot g'),$$

and

$$\begin{aligned} U(g)U(g')\Phi_i &= U(g)\sum_j \Phi_j D_{ji}(g') = \sum_j U(g)\Phi_j D_{ji}(g') \\ &= \sum_j \left(\sum_k \Phi_k D_{kj}(g)\right) D_{ji}(g') = \sum_k \Phi_k \left(\sum_j D_{kj}(g) D_{ji}(g')\right). \end{aligned}$$

Now if we put together the above two equations, we can prove the first relation in (1.2.4):

$$D_{ji}(g \cdot g') = \sum_k D_{kj}(g) D_{ji}(g'),$$

which is the multiplication rule for the matrices $D(g)$.

The second relation in (1.2.4) is a consequence of (1.2.3) and the orthonormality of the basis $\{\Phi_i\}$:

$$\begin{aligned} \delta_{ij} &= \langle \Phi_i | \Phi_j \rangle = \langle U(g)\Phi_i | U(g)\Phi_j \rangle \\ &= \left\langle \sum_k \Phi_k D_{ki}(g) \middle| \sum_l \Phi_l D_{lj}(g) \right\rangle = \sum_{k,l} D_{ki}(g)^* \langle \Phi_k | \Phi_l \rangle D_{lj}(g) \\ &= \sum_k D_{ik}(g)^\dagger D_{kj}(g) = (D(g)^\dagger D(g))_{ij}, \end{aligned}$$

where in the last steps we have transposed the indices and used the definition of the adjoint matrix $D(g)^\dagger$. This proves that $D(g)^\dagger D(g) = \mathbb{I}$. $\square$

Having proven (1.2.4), we know $g \mapsto D(g)$ to be a unitary representation of the symmetry group G on the eigenspace $\mathcal{H}_E$ corresponding to the energy eigenvalue E, and from (1.2.3) we know the basis vectors Φ_i to transform under the action of G through $D(g)$.

The energy eigenvalue E can thus be classified according to the unitary representation $D(g)$, where the energy eigenstates in $\mathcal{H}_E$ transform under the action of G. The problem is that there are infinitely many orthonormal bases of the energy eigenspace $\mathcal{H}_E$ of E. The representation $D(g)$ depends on the chosen basis $\{\Phi_i\}$. So $D(g)$ by itself cannot be used to distinguish E from other energy eigenvalues. Let us analyze this point in greater detail.

Proposition 1.7. *Let $\{\Phi^1{}_i\}$ and $\{\Phi^2{}_i\}$ be two ON bases for the eigenspace $\mathcal{H}_E$ corresponding to the energy eigenvalue E. Then, let the corresponding unitary representations be $D^1(g)$ and $D^2(g)$, defined by* (1.2.3) *with respect to the bases $\{\Phi^1{}_i\}$ and $\{\Phi^2{}_i\}$, respectively.*

There exist a $d_E \times d_E$ unitary matrix A such that its elements A_{ij} relate the two bases $\{\Phi^1{}_i\}$ and $\{\Phi^2{}_i\}$ as

$$\Phi^2{}_i = \sum_{j=1}^{d_E} \Phi^1{}_j A_{ji}, \tag{1.2.5}$$

and the two unitary representations $D^1(g)$ and $D^2(g)$ are related by the similarity transformation

$$D^2(g) = A^{-1} D^1(g) A, \tag{1.2.6}$$

for all g in G. Thus, the two unitary representations $D^1(g)$ and $D^2(g)$ are equivalent representations of the symmetry group G.

Proof. Using (1.2.3), we can develop (1.2.5) as

$$\begin{aligned} U(g)\Phi^2{}_i = U(g)\sum_{j=1}^{d_E} \Phi^1{}_j A_{ji} = \sum_{j=1}^{d_E} U(g)\Phi^1{}_j A_{ji} = \sum_{j=1}^{d_E} \left(\sum_{k=1}^{d_E} \Phi^1{}_k D^1_{kj}(g) \right) A_{ji} \\ = \sum_{k=1}^{d_E} \Phi^1{}_k \left(\sum_{j=1}^{d_E} D^1_{kj}(g) A_{ji} \right) = \sum_{k=1}^{d_E} \Phi^1{}_k (D^1(g)A)_{ki}, \end{aligned}$$

on one hand, and

$$\begin{aligned} U(g)\Phi^2{}_i = \sum_{k=1}^{d_E} \Phi^2{}_k D^2_{ki}(g) = \sum_{k=1}^{d_E} \left(\sum_{j=1}^{d_E} \Phi^1{}_j A_{jk} \right) D^2_{ki}(g) \\ = \sum_{j=1}^{d_E} \Phi^1{}_j \left(\sum_{k=1}^{d_E} A_{jk} D^2_{ki}(g) \right) = \sum_{j=1}^{d_E} \Phi^1{}_j (AD^2(g))_{ji}. \end{aligned}$$

Putting together the above two equations, we have

$$(D^1(g)A)_{ki} = (AD^2(g))_{ki}, \quad \text{i.e., } D^1(g)A = AD^2(g),$$

which shows that the two unitary representations $D^1(g)$ and $D^2(g)$ are related by the similarity transformation $D^2(g) = A^{-1}D^1(g)A$. This proves that the two unitary representations $D^1(g)$ and $D^2(g)$ are equivalent representations of the symmetry group G. □

Thus two unitary representations $D^1(g)$ and $D^2(g)$ of the symmetry group G of the same dimension d_E, related as in (1.2.6), for some unitary matrix A, are said to be **unitarily equivalent**. Thus the unitary representation $D(g)$ in (1.2.3) is independent of the choice of the ON basis $\{\Phi_i\}$ for the eigenspace $\mathcal{H}_E$ up to unitary equivalence.

We can in this way characterize the energy eigenvalue E by a *unitary equivalence class of unitary representations*[2] of G and use this class to group E with other eigenvalues sharing the very same class. In this way, **a symmetry based classification of the energy spectrum is achieved**. Since however there are infinitely many representations of any given group, the classification scheme obtained may not be so useful. Fortunately, generic representations can be decomposed as combinations of a smaller set of more basic representations, which can then be used to refine the scheme.

Reducibility and Irreducibility

Suppose that we are able to chose the orthonormal basis $\{\Phi_i\}$ of the energy eigenspace $\mathcal{H}_E$ to be of indexed $\Phi_{\alpha r}$ by two indices α, r in such a way that (1.2.3) takes the form

$$U(g)\Phi_{\alpha r} = \sum_s \Phi_{\alpha s} D_{\alpha s r}(g),$$

for every element g of G, for certain g-dependent coefficients $D_{\alpha s r}(g)$. Therefore, the vectors $\Phi_{\alpha r}$ for fixed α mix among themselves under the action of G. This means that, for fixed g, the matrix $D(g)$ has the block diagonal form

$$D(g) = \begin{pmatrix} D_1(g) & 0 & \cdots & 0 \\ 0 & D_2(g) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & D_a(g) \end{pmatrix},$$

and let us assume that the blocks $D_\alpha(g)$ are of dimension d_α, which is the smallest non trivial dimension for each block (i.e. the blocks cannot be further decomposed into diagonal blocks by a furhter change of basis $\Phi_{\alpha r}$). Then, the unitary representation $g \mapsto D(g)$ of G on $\mathcal{H}_E$ is said to be **reducible**, and its blocks $D_\alpha(g)$ are said to define an **irreducible** representations $g \mapsto D_\alpha(g)$ of G (*irreducible components*).

The energy eigenvalue E can then be classified through the irreducible representations $D_\alpha(g)$ of G. The former way of proceeding provides a classification scheme of the energy spectrum relying on a more restricted set of elements and so potentially more useful than that furnished by the latter.

We can resume the results obtained in this section as follows:

- Each energy eigenvalue E is characterized by a set of irreducible unitary representations D_α of G, unique up to unitary equivalence, describing how the eigenvectors belonging to E transform under the action of G.
- The energy eigenvalues can be grouped in subsets of eigenvalues sharing the same representation content D_α providing a classification scheme of the energy spectrum.

Let us illustrate the above results with a few examples.

Example (The One-Dimensional Harmonic Oscillator). The parity transformation P is space reflection about the origin. Observation show that the parity group $C_2 = \{\mathbb{I}, P\}$ is a dynamical symmetry group of the one–dimensional harmonic oscillator.

[2]It is a class of equivalence of unitary representations, where the representations are unitarily equivalent.

As well known, the energy eigenvalues of the one–dimensional harmonic oscillator are given by

$$H\Phi_n = E_n\Phi_n. \quad E_n = \hbar\omega(n+1/2),$$

with $n = 0, 1, 2, \ldots$, and are non degenerate. The energy eigenstates of the one–dimensional harmonic oscillator transform under the action of the parity transformation P as

$$U(P)\Phi_n = \Phi_n(-1)^n.$$

Thus the parity group C_2 has two irreducible unitary representations: the trivial representation $D_+(P) = 1$ and the sign representation $D_-(P) = -1$. The energy eigenvalues of the one–dimensional harmonic oscillator are classified according to these two irreducible unitary representations of C_2: the energy eigenvalues with even n transform according to the trivial representation D_+, while those with odd n transform according to the sign representation D_-. Thus we classify them as even or odd energy eigenvalues according to the parity of n.

Example (The Hydrogen Atom). It can be shown that the proper rotation group SO(3) is a dynamical symmetry group of the hydrogen atom. The energy eigenvalues of the hydrogen atom are known to be given by

$$H\Phi_{nlm} = E_n\Phi_{nlm}, \quad E_n = -\frac{mc^2\alpha^2}{2n^2}.$$

The energy eigenvalues of the hydrogen atom are degenerate, with degeneracy $d_{E_n} = n^2$:

1. $n = 0, 1, 2, \ldots,$
2. $l = 0, 1, \ldots, n-1,$
3. $m = -l, -l+1, \ldots, l-1, l,$

and the eigenstates Φ_{nlm} are also eigenstates of the total angular momentum operator L^2 and of the component along the quantization axis of the angular momentum operator L_z:

$$\begin{cases} L^2\Phi_{nlm} = \hbar^2 l(l+1)\Phi_{nlm}, \\ L_0\Phi_{nlm} = \hbar m\Phi_{nlm}. \end{cases}$$

Furthermore, the other spherical coordinates of the angular momentum $L_\pm$ act on the eigenstates Φ_{nlm} as

$$L_\pm\Phi_{nlm} = \hbar\sqrt{l(l+1) - m(m\pm1)}\,\Phi_{nl(m\pm1)},$$

thus acting as ladder operators on the index m.

Since $\mathbf{L}$ is the **infinitesimal generator** of the proper rotation group SO(3) (which will be discussed in the next section 1.3), the energy eigenstates Φ_{nlm} transform under the action of SO(3) according to the irreducible unitary representation $D_l(\mathbf{R})$ of SO(3), of dimension $d_l = 2l+1$:

$$U(\mathbf{R})\Phi_{nlm} = \sum_{m'=-l}^{l} \Phi_{nlm'} D_{lm'm}(\mathbf{R}),$$

so that $D_l(\mathbf{R})$ is $(2l+1)\times(2l+1)$ dimensional, and the mappings $\mathbf{R} \mapsto D_l(\mathbf{R})$ can be shown to be unitary irreducible representations of SO(3). Thus for each energy eigenvalue E_n of the hydrogen atom, the energy eigenstates Φ_{nlm} transform under the action of SO(3) according to the n irreducible $(2l+1)$-dimensional unitary representation $D_{l=0,1,\ldots,n-1}$ of SO(3). In this way, the symmetry based classification of the energy spectrum is achieved for the hydrogen atom.

TODO: put image of the molecule

Example (The Sulfur Trioxide Molecule SO_3). The molecule of sulfur trioxide SO_3 has the shape of a planar equilateral triangle, with the sulfur atom at the center and the three oxygen atoms at the vertices. The symmetry group of this molecule is the dihedral group D_3, which is a subgroup of SO(3).

A planar equilateral triangle has the following symmetries:

- Three rotations of 120° about the axis perpendicular to the plane of the triangle, which we call C_3 rotations:

$$0^\circ : \begin{pmatrix} 123 \\ \downarrow \\ 123 \end{pmatrix}, \quad 120^\circ : \begin{pmatrix} 123 \\ \downarrow \\ 231 \end{pmatrix}, \quad 240^\circ : \begin{pmatrix} 123 \\ \downarrow \\ 312 \end{pmatrix}.$$

- Three reflections about the axes of symmetry of the triangle, which we call σ_v reflections:

$$\sigma_v^1 : \begin{pmatrix} 123 \\ \downarrow \\ 132 \end{pmatrix}, \quad \sigma_v^2 : \begin{pmatrix} 123 \\ \downarrow \\ 321 \end{pmatrix}, \quad \sigma_v^3 : \begin{pmatrix} 123 \\ \downarrow \\ 213 \end{pmatrix}.$$

Putting together the above symmetries, we find a total of six symmetries, which form the dihedral group D_3, which coincides with the symmetric group S_3 of permutations of three objects. The energy eigenvalues of the sulfur trioxide molecule SO_3 can be classified according to the irreducible unitary representations of D_3.

This group has three *unitary equivalence classes of irreducible unitary representations*: the trivial representation D_+, the sign representation D_- and a two-dimensional representation D_2. They are of dimensions $d_+ = 1$, $d_- = 1$ and $d_2 = 2$, respectively, and we can deduce their dimensionality through the following theorem:

Theorem 1.8. *The sum of the squares of the dimensions d_α of the irreducible unitary representations D_α of a finite non-abelian group G is equal to the order $|G|$ (the number of elements) of the group:*

$$\sum_\alpha d_\alpha^2 = |G|.$$

In our case the only way to satisfy the above relation is to have $d_+^2 + d_-^2 + d_2^2 = 1^2 + 1^2 + 2^2 = 6$, which is the order of D_3. The energy eigenvalues of the sulfur trioxide molecule SO_3 can be classified according to these three irreducible unitary representations of D_3.

1.3 | Lie Groups and Lie Algebras

A Lie group G is an infinite group that can be parametrized in a neighborhood of the neutral element by continuous parameters θ^a, $a = 1, \dots, d$, organized as a vector $\underline{\theta}$, in one–to–one fashion:

$$\begin{cases} \underline{\theta} \in D \subseteq \mathbb{R}^d, \\ g \,:\, D \mapsto G, \\ \underline{\theta} \mapsto g(\underline{\theta}), \end{cases}$$

where D is an open neighborhood of the origin in $\mathbb{R}^d$ ($\underline{0} \in D$ and $g(\underline{0}) \in G$). The number d of parameters is called the dimension of G, seen as a differential map. The group operations of multiplication and inversion are smooth functions of the parameters.

If the parametrization is smooth, as we assume, G can be thought geometrically as a group manifold. Below, we treat G as a matrix group for definiteness.

We assume that 0 belongs to the parameter space and that

$$g(\underline{0}) = 1,$$

which can be read as a normalizing condition. By virtue of the parametrization, for any pair of parameter vectors $\underline{\theta}$, $\underline{\theta}'$, there exists a third parameter vector $\underline{\theta}''$ such that

$$g(\underline{\theta}')g(\underline{\theta}) = g(\underline{\theta}'').$$

As the parametrization is one–to–one,[3] $\underline{\theta}''$ is also uniquely determined by $\underline{\theta}$, $\underline{\theta}'$. So, it can be expressed as a function of them

$$\underline{\theta}'' = \underline{m}(\underline{\theta}', \underline{\theta}),$$

which is a function of the **group multiplication**. The function $\underline{m}$ is smooth as the group multiplication is smooth. The inverse of $g(\underline{\theta})$ is given by

$$g(\underline{\theta})^{-1} = g(\underline{\theta}'),$$

for some $\underline{\theta}' = \underline{j}(\underline{\theta})$, which is unique since the parameterization is one-to-one. The function $\underline{j}$, implementing the group inversion, is smooth as the group inversion is smooth.

We have thus found two smooth functions $\underline{m}$, $\underline{j}$ that encode the group multiplication and inversion in terms of the parameters $\underline{\theta}$:

$$\begin{cases} \underline{m} : D \times D \to D, \\ \underline{j} : D \to D, \end{cases} \quad \text{such that} \quad \begin{cases} g(\underline{\theta}')g(\underline{\theta}) = g(\underline{m}(\underline{\theta}', \underline{\theta})), \\ g(\underline{j}(\underline{\theta})) = g(\underline{\theta})^{-1}. \end{cases} \tag{1.3.1}$$

The group properties of associativity, invertibility and neutrality, which are satisfied by the group multiplication and inversion, translate into properties of the functions $\underline{m}$, $\underline{j}$ as follows:

Proposition 1.9. *The following relations must hold:*

$$\begin{aligned} m(m(\underline{\theta}, \underline{\theta}'), \underline{\theta}'') &= m(\underline{\theta}, m(\underline{\theta}', \underline{\theta}'')), \\ m(\underline{\theta}, j(\underline{\theta})) &= m(j(\underline{\theta}), \underline{\theta}) = \underline{0}, \\ m(\underline{\theta}, \underline{0}) &= m(\underline{0}, \underline{\theta}) = \underline{\theta}. \end{aligned} \tag{1.3.2}$$

Given any choice of three parameter vectors $\underline{\theta}$, $\underline{\theta}'$, $\underline{\theta}''$. These identities constrain the form of the group multiplication function m and the inversion function j to an important extent.

[3] This means that each element of the group corresponds to a unique set of parameters: $g(\underline{\theta}) = g(\underline{\theta}') \implies \underline{\theta} = \underline{\theta}'$.

Proof. The first identity is a consequence of the associativity of the group multiplication: using the definition of the multiplication function m along with the associativity of the group multiplication, we find

$$\begin{aligned} g(m(m(\underline{\theta},\underline{\theta}'),\underline{\theta}'')) &= g(m(\underline{\theta},\underline{\theta}'))g(\underline{\theta}'') = g(\underline{\theta})g(\underline{\theta}')g(\underline{\theta}'') \\ &= g(\underline{\theta})g(m(\underline{\theta}',\underline{\theta}'')) = g(m(\underline{\theta},m(\underline{\theta}',\underline{\theta}''))), \end{aligned}$$

which implies the first identity in $(1.3.2)_1$ by the one-to-one nature of the parametrization (otherwise we would have $g(\underline{\theta}) = g(\underline{\theta}')$ for some $\underline{\theta} \neq \underline{\theta}'$). The second identity is a consequence of the invertibility of the group multiplication: using the definition of the inversion function j along with the invertibility of the group multiplication, we find

$$\begin{aligned} g(m(\underline{\theta},j(\underline{\theta}))) &= g(\underline{\theta})g(j(\underline{\theta})) = g(\underline{\theta})g(\underline{\theta})^{-1} = 1 = g(\underline{0}), \\ g(m(j(\underline{\theta}),\underline{\theta})) &= g(j(\underline{\theta}))g(\underline{\theta}) = g(\underline{\theta})^{-1}g(\underline{\theta}) = 1 = g(\underline{0}), \end{aligned}$$

proving both sides of the second identity in $(1.3.2)_2$. The third identity is a consequence of the neutrality of the group multiplication: using the definition of the multiplication function m along with the neutrality of the group multiplication, we find

$$g(m(\underline{0},\underline{\theta})) = g(\underline{\theta}), \quad g(m(\underline{\theta},\underline{0})) = g(\underline{\theta}),$$

proving both sides of the third identity in $(1.3.2)_3$. □

1.3.1 | Local Structure of the Group

The idea now is to understand the structure of G by looking at the behavior of $g(\underline{\theta})$ in a neighborhood of the neutral element. This is done by Taylor expanding in power series around $\underline{0}$ as:

$$g(\underline{\theta}) = g(\underline{0}) + \left.\frac{\partial g}{\partial \theta^a}\right|_{\underline{0}} \theta^a + \frac{1}{2}\left.\frac{\partial^2 g}{\partial \theta^a \partial \theta^b}\right|_{\underline{0}} \theta^a\theta^b + \mathrm{O}(\theta^3).$$

Now computing individual terms of the expansion, we have

$$g(\underline{0}) = 1, \quad t_a = \left.\frac{\partial g}{\partial \theta^a}\right|_{\underline{0}}, \quad t_{ab} = \left.\frac{\partial^2 g}{\partial \theta^a \partial \theta^b}\right|_{\underline{0}} = t_{ba},$$

so that

$$g(\underline{\theta}) = 1 + t_a\theta^a + \frac{1}{2}t_{ab}\theta^a\theta^b + \mathrm{O}(\theta^3). \tag{1.3.3}$$

Let us highlight the fact that the coefficients t_a, t_{ab} are matrices, as $g(\underline{\theta})$ is a matrix group. t_{ab} in particular is symmetric in its indices a, b since the order of differentiation does not matter.

We can now think of G as an **hypersurface in the space of matrices**, containing the identity matrix. In the neighborhood of the identity, there is D, subset of our hypersurface, which is diffeomorphic to an open neighborhood of the origin in $\mathbb{R}^d$ (the domain of our parametrization).

If we vary $g(0,\dots,0,\theta^a,0,\dots,0)$, we get a curve in D passing through the identity (when $\theta^a = 0$). The tangent vector to this curve at the identity is given by $t_a = \frac{\partial g}{\partial \theta^a}\big|_{\underline{0}}$. Varying all parameters θ^a we get d curves in D passing through the identity and their tangent vectors at the identity are given by t_a, $a = 1,\dots,d$. These tangent vectors are **linearly independent** since the parametrization is one–to–one, thus they span a vector space of dimension d.

The tangent space to G at the identity is more than a vector space of dimension d (spanned by the matrices t_a, $a = 1,\dots,d$, costituing a basis). We denote this tangent space by $\mathfrak{g}$ and call it the

Lie algebra of G. It is a vector space of dimension d and it is closed under the commutator of matrices: the Lie algebra $\mathfrak{g}$ encodes all the information about the local structure of G around the identity element, but this will be clarified in a moment.

We could now think to Taylor expand the group multiplication functions m, j around $\underline{0}$ and try to find relations among the coefficients of the expansion which encode the group structure. Starting from the multiplication function, we find that the coefficients of the expansion of m must satisfy

$$m^a(\underline{\theta}, \underline{\theta}') = \theta^a + \theta'^a + m^a{}_{bc}\theta^b\theta'^c + \mathrm{O}(\underline{\theta}^3), \tag{1.3.4}$$

while for the inversion property we find

$$j^a(\underline{\theta}) = -\theta^a + j^a{}_{bc}\theta^b\theta^c + \mathrm{O}(\underline{\theta}^3), \tag{1.3.5}$$

where $m^a{}_{bc}$, $j^a{}_{bc}$ are the coefficients of the expansions. These coefficients are not independent, as it can be proven that they satisfy:

$$j^a{}_{bc} = m^a{}_{bc} + m^a{}_{cb}.$$

If we now recall the Taylor expansion of $g(\underline{\theta})$ around $\underline{0}$

$$g(\underline{0}) = 1 + t_a\theta^a + \frac{1}{2}t_{ab}\theta^a\theta^b + \mathrm{O}(\theta^3),$$

we can note how the coefficients t_a define another member of the Lie algebra $\mathfrak{g}$: the Lie algebra $\mathfrak{g}$ is the tangent space to G at the identity, so it is spanned by the matrices t_a, $a = 1, \ldots, d$, which we have said to be closed under the commutator of matrices. Let us now define the commutator of two basis elements t_a, t_b of $\mathfrak{g}$ as follows:

Definition 1.1 (Lie Bracket)**.** The **commutator**, or **Lie bracket**, of two basis elements t_a, t_b of $\mathfrak{g}$ is defined as follows:

$$[t_a, t_b] = t_a t_b - t_b t_a = f_{ab}{}^c t_c \in \mathfrak{g},$$

where $f_{ab}{}^c$ are the **structure constants** of the Lie algebra $\mathfrak{g}$, encoding the properties of the Lie algebra $\mathfrak{g}$ and thus of the local structure of the Lie group G around the identity element.

Having defined the Lie bracket of two basis elements t_a, t_b of $\mathfrak{g}$, we can now prove the following proposition:

Proposition 1.10. *The structure constants are antisymmetric in their lower indices a, b since the commutator is antisymmetric. These structure constants are related to the coefficients $m^a{}_{bc}$ of the expansion of the group multiplication function by*

$$f^a{}_{bc} = m^a{}_{bc} - m^a{}_{cb}.$$

If we change parametrization, the structure constants change accordingly, but the Lie algebra $\mathfrak{g}$ remains the same.

Proof. Taking again the definition of the multiplication function

$$g(\underline{\theta}')g(\underline{\theta}) = g(m(\underline{\theta}', \underline{\theta})),$$

we can Taylor expand both sides around $\underline{0}$ and find the following relation between the coefficients of the expansion of g and the structure constants:

$$\left[1 + \theta^a t_a + \frac{1}{2}\theta^a\theta^b t_{ab} + \mathrm{O}(\theta^3)\right]\left[1 + \theta'^a t_a + \frac{1}{2}\theta'^a\theta'^b t_{ab} + \mathrm{O}(\theta'^3)\right]$$
$$= 1 + \left(\theta^a + \theta'^a + m^a{}_{bc}\theta^b\theta'^c\right)t_a + \frac{1}{2}\left(\theta^a + \theta'^a\right)\left(\theta^b + \theta'^b\right)t_{ab} + \mathrm{O}(\theta^2, \theta') + \mathrm{O}(\theta, \theta'^2).$$

Relating the coefficients of the expansion of both sides in $\theta^a\theta'^b$, we find the following relations (renaming summation indices):

$$t_a t_b = t_{ab} + m^c{}_{ab} t_c,$$
$$t_b t_a = t_{ba} + m^c{}_{ba} t_c,$$

but since $t_{ab} = t_{ba}$, if we subtract the second from the first, we find

$$[t_a, t_b] = (m^c{}_{ab} - m^c{}_{ba}) t_c = f_{ab}{}^c t_c,$$

so that the structure constants $f_{ab}{}^c$ of the Lie algebra $\mathfrak{g}$ are related to the coefficients $m^c{}_{ab}$ of the expansion of the group multiplication function m by

$$f^c{}_{ab} = m^c{}_{ab} - m^c{}_{ba}.$$

□

Thus to sum up:

- The Lie algebra $\mathfrak{g}$ of a Lie group G is the tangent space to G at the identity element, spanned by the matrices t_a, $a = 1, \ldots, d$.
- The basis elements t_a of the Lie algebra $\mathfrak{g}$ satisfy the commutation relations

  $$[t_a, t_b] = f_{ab}{}^c t_c,$$

  where $f_{ab}{}^c$ are the structure constants of the Lie algebra $\mathfrak{g}$.
- The structure constants $f_{ab}{}^c$ of the Lie algebra $\mathfrak{g}$ are related to the coefficients $m^c{}_{ab}$ of the expansion of the group multiplication function m by

  $$f^c{}_{ab} = m^c{}_{ab} - m^c{}_{ba}.$$

The Lie algebra, the tangent space to the Lie group at the identity, is not only a vector space, but also an algebra, since it is closed under the commutator of matrices. It encodes all the information about the local structure of the Lie group around the identity element. In particular, the structure constants $f_{ab}{}^c$ determine the commutation relations of the basis elements t_a of the Lie algebra, which in turn determine the local structure of the Lie group G around the identity element.

The commutator of the basis elements t_a of the Lie algebra $\mathfrak{g}$ has some properties, which are the **Jacobi identity** and the **antisymmetry** of the commutator:

$$\begin{cases} [t_a,\, t_b] + [t_b,\, t_a] = 0, \\ [t_a,\, [t_b,\, t_c]] + [t_b,\, [t_c,\, t_a]] + [t_c,\, [t_a,\, t_b]] = 0. \end{cases}$$

These two properties can be expressed in terms of the structure constants $f_{ab}{}^c$ as follows:

$$\begin{cases} f_{ab}{}^c + f_{ba}{}^c = 0, \\ f_{ad}{}^e f_{bc}{}^d + f_{bd}{}^e f_{ca}{}^d + f_{cd}{}^e f_{ab}{}^d = 0, \end{cases}$$

which are the **antisymmetry** of the structure constants in their lower indices and the **Jacobi identity** for the structure constants, respectively. This means that not every set of numbers $f_{ab}{}^c$ can be the structure constants of a Lie algebra, but only those that satisfy the antisymmetry and Jacobi identity properties.

1.3.2 | Exponential Parametrization of Lie Groups

Given the parametrization of a Lie group G in a neighborhood of the identity element

$$g \,:\, D \mapsto G, \quad \underline{\theta} \in D \mapsto g(\underline{\theta}) \in G,$$

is it possible to find a global parametrization of G in terms of the elements of its Lie algebra $\mathfrak{g}$? The answer is yes, and the way to do it is through the *exponential map*.

If we consider another parametrization of G in a neighborhood of the identity element defined by

$$\tilde{g}(\underline{\theta}) = \lim_{N\to\infty} g\left(\tfrac{\underline{\theta}}{N}\right)^N,$$

which if the group is suitably closed under the group multiplication, is well defined and provides a global parametrization of G in terms of the elements of its Lie algebra $\mathfrak{g}$:

$$\tilde{g}(\underline{\theta}) = e^{\theta^a t_a}.$$

This new parametrization is called the **exponential parametrization** of G and the map $\tilde{g} : \mathbb{R}^d \to G$ is called the **exponential map**.

Proof. Let us show that the limit defining $\tilde{g}(\underline{\theta})$ exists and is equal to $e^{\theta^a t_a}$. Starting from the definition of $\tilde{g}(\underline{\theta})$, we can Taylor expand $g\left(\underline{\theta}/N\right)$ around $\underline{0}$ as follows:

$$g\left(\frac{\theta}{N}\right)^N = \left[1 + \frac{t_a\theta^a}{N} + \frac{1}{2}\left(\frac{t_a\theta^a}{N}\right)^2 + \mathrm{O}\left(\frac{1}{N^3}\right)\right]^N \sim \left(1 + \frac{t_a\theta^a}{N}\right)^N \xrightarrow{N\to\infty} e^{t_a\theta^a},$$

which is the exponential of a matrix, defined by Euler's formula as the power series expansion

$$e^x = \lim_{n\to\infty}(1+\frac{x}{n})^n = \sum_{k=0}^{\infty}\frac{x^k}{k!}.$$

□

Thus, the exponential map provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ as follows:

$$\tilde{g}(\underline{\theta}) = e^{t_a\theta^a} = 1 + \tilde{t}_a\theta^a + \frac{1}{2}\theta^a\theta^b\tilde{t}_{ab} + \mathrm{O}(\theta^3),$$

which is the same expansion did in (1.3.3). But when comparing the coefficients to the ones of the expansion of $e^{i\theta^a t_a}$, we get the following relations:

$$\tilde{t}_a = t_a, \quad \tilde{t}_{ab} = \tfrac{1}{2}t_a t_b + t_b t_a.$$

The same goes for the he group multiplication function $\underline{m}$ and the inversion function $\underline{j}$. We can define new functions $\underline{\tilde{m}}$, $\underline{\tilde{j}}$ as

$$\tilde{g}(\underline{\tilde{m}}(\underline{\theta},\underline{\theta}')) = \tilde{g}(\underline{\theta}')\tilde{g}(\underline{\theta}),$$
$$\tilde{g}(\underline{\tilde{j}}(\underline{\theta})) = \tilde{g}(\underline{\theta})^{-1},$$

which can be thought as the group multiplication and inversion functions in the exponential parametrization. If we now Taylor expand these new functions around $\underline{0}$ we find that for the inversion function it is as the previous expansion (1.3.5), but all the orders higher than the first vanish:

$$\tilde{j}(\underline{\theta}) = -\theta^a, \quad \tilde{j}^a{}_{bc} = 0$$
$$e^{\underline{\theta}^a t_a} = \tilde{g}(\tilde{j}(\underline{\theta})) = \tilde{g}(\underline{\theta})^{-1} = e^{-t_a\theta^a}.$$

For the multiplication function we have instead the **Campbell-Baker-Hausdorff function** so that the expansion of the multiplication function in the exponential parametrization is given by

$$\tilde{m}^a(\underline{\theta}, \underline{\theta}') = \theta^a + \theta'^a + \tilde{m}^a{}_{bc}\theta^b\theta'^c + \mathrm{O}(\underline{\theta}^3),$$

which is the same of (1.3.4) but with different coefficients $\tilde{m}^a{}_{bc}$ of the expansion, which are related to the structure constants $f^a_{\ bc}$ of the Lie algebra $\mathfrak{g}$ by

$$\tilde{m}^a{}_{bc} = \tfrac{1}{2} f^a{}_{bc}.$$

These new structure constants are not different from the original ones, since the exponential map is a reparametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$: $\tilde{t}_a = t_a$ implies that

$$f^c{}_{ab}t_c = [t_a,\, t_b] = \left[\tilde{t}_a,\, \tilde{t}_b\right] = \tilde{f}^c{}_{ab}\tilde{t}_c = \tilde{f}^c{}_{ab}t_c,$$

thus the structure constants of the Lie algebra $\mathfrak{g}$ are the same in both parametrizations, as expected since the Lie algebra is an intrinsic property of the Lie group G and does not depend on the choice of parametrization:

$$f^c{}_{ab} = \tilde{f}^c{}_{ab}.$$

Thus we have

$$\tilde{m}^a{}_{bc} = \frac{1}{2}\tilde{f}^a{}_{bc} = \frac{1}{2} f^a{}_{bc},$$

and if we expand the CBH function in power series we find[4]

$$\tilde{m}^a(\theta, \theta') = \theta^a + \theta'^a + \frac{1}{2} f^a{}_{bc}\theta^b\theta'^c + \mathrm{O}(\theta^3, \theta'^3),$$

which is like a polynomial expansion of the group multiplication function m in terms of the structure constants $f^a{}_{bc}$ of the Lie algebra $\mathfrak{g}$. The exponential map thus provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ and allows us to express the group multiplication function m in terms of the structure constants $f^a{}_{bc}$ of the Lie algebra $\mathfrak{g}$.

Example (Phase Group U(1)). Let us define the group of phase transformations U(1) as the group of complex numbers of unit modulus under multiplication:

$$\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\}.$$

It is an abelian Lie group of dimension 1 which represents the Gauge group of QED. The elements of U(1) can be represented by an angle θ with an exponential parametrization as follows:

$$\tilde{g}(\theta) = e^{i\theta^a t_a} = e^{i\theta},$$

since the Lie algebra of U(1) is one-dimensional and spanned by the identity matrix $t = 1$. The structure constants of the Lie algebra of U(1) are zero since the commutator of any two elements of the Lie algebra is zero:

$$[t, t] = 0 \implies f_{ab}{}^c = 0.$$

[4]There exists a closed form expression for the CBH function, but it is not important for our purposes: remind only that exact complicated expressions for computing all the teerms exist, but we are only interested in the first few terms of the expansion.

Example (Special Unitary Group SU(2)). The special unitary group SU(2) is the group of 2×2 unitary matrices with determinant 1:

$$\mathrm{SU}(2) = \{U \in \mathbb{C}^{2\times 2} : U^\dagger U = \mathbb{I}_2, \det(U) = 1\}.$$

It is a non-abelian Lie group of dimension 3: $\underline{\theta} \in \mathbb{R}^3$. The elements of SU(2) can be parametrized using the Pauli matrices σ_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i \frac{\tau(\theta)}{2}},$$

where

$$\tau(\theta) = \begin{pmatrix} \theta^3 & \theta^1 - i\theta^2 \\ \theta^1 + i\theta^2 & -\theta^3 \end{pmatrix} = \theta^1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \theta^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \theta^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \theta^i \sigma_i,$$

Thus $t_a = \frac{1}{2}\tau_a$ and the structure constants of the Lie algebra $\mathfrak{su}(2)$ are given by the Levi-Civita symbol ϵ_{ijk}:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \implies f_{ij}{}^k = 2\epsilon_{ijk}.$$

Example (Special Orthogonal Group SO(3)). The special orthogonal group SO(3) is the group of 3×3 real orthogonal matrices with determinant 1:

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = \mathbb{I}_3, \det(R) = 1\}.$$

It is a non-abelian Lie group of dimension 3. The elements of SO(3) can be parametrized using the generators of rotations J_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i J_i},$$

where the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk} J_k.$$

[-....-] to be finished

1.4 | Unitary Representations of Continuous Symmetries

If the symmetry group G is a Lie group, then the quantum unitary operators representing the group action on states and observables can be expressed as functions of the parameter vector $\underline{\theta}$ of G:

[...]

$$U(m(\theta,\theta')) = U(g(m(\theta,\theta'))) = U(g(\theta)g(\theta')) = \gamma(g(\theta),g(\theta'))U(g(\theta))U(g(\theta')),$$

where $\gamma(g(\theta),g(\theta'))$ is a phase factor which can be written as $\gamma(g(\theta),g(\theta')) = e^{i\omega(g(\theta),g(\theta'))}$ for some real function ω. Thus

$$U(m(\theta,\theta')) = e^{i\omega(g(\theta),g(\theta'))}U(g(\theta))U(g(\theta')).$$

Example (EXCERCISE).

??? photo

When you have a Lie Group and a projected unitary representation in the Hilbert space of states, you can always get to this relation with the phase factor γ by redefining the unitary operators $U(g)$ with a suitable phase factor, all given to the properties of the Lie group G and the properties of the projective representation U.

Since the Lie groups act homogeneously, we can always translates what happens at the identity element to any other element of the group. Thus, by looking at the properties of the group around the identity element, we can understand the properties of the whole group. By Taylor expanding $U(\underline{\theta})$ near the neutral element $\underline{0}$ of the group, we get

$$U(\underline{\theta}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a\theta^b Q_{ab} + \mathrm{O}(\theta^3), \quad Q_{ab} = Q_{ba},$$

where Q_a are the generators of the representation U of the Lie group G. The generators Q_a are essentially the tangent vectors to the representation U of the Lie group G at the identity element

$$Q_a = i\frac{\partial U(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}}.$$

We are now going to prove that the generators Q_a of the representation U of the Lie group G are self-adjoint operators:

$$Q_a^\dagger = Q_a.$$

We can start from the adjoint

$$\begin{aligned}
Q_a^\dagger &= \left(i\frac{\partial U(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}}\right)^\dagger \\
&= -i\frac{\partial U^\dagger(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}} \\
&= -i\frac{\partial U^{-1}(\underline{\theta})}{\partial\theta^a}\Big|_{\underline{\theta}=\underline{0}} \\
&= +iU(\underline{\theta})^{-1}\frac{\partial U(\underline{\theta})}{\partial\theta^a}U(\underline{\theta})^{-1}\Big|_{\underline{\theta}=\underline{0}} = Q_a,
\end{aligned}$$

where we have used the unitarity of U and the fact that $U(\underline{0}) = 1$. [comments]

We have another property to prove: we have to normalize the phase function ω

$$\omega(\underline{\theta}, \underline{0}) = \omega(\underline{0}, \underline{\theta}) = 0,$$

so that we remember that

$$\omega(\underline{\theta}, \underline{\theta}') = \theta^a \theta'^b \omega_{ab} + \mathrm{O}(\theta^3),$$

so that the phase factor γ is normalized to 1 at the identity element of the group, since

$$e^{i\omega(\underline{0},\underline{\theta})} = \gamma(g(\underline{0}), g(\underline{\theta}')) = 1 = e^{i\omega(\underline{\theta},\underline{0})} = \gamma(g(\underline{\theta}), g(\underline{0})) = 1.$$

???

[...]

$$[Q_a, Q_b]$$

... as we did for $g(\theta)g(\theta')$ we can multiply the taylor expansion of $U(\theta)$ and $U(\theta')$ and compare the result with the taylor expansion of $U(m(\theta, \theta'))$ to get

$$U(\underline{\theta})U(\boldsymbol{\theta'}) = e^{-i\omega(\underline{\theta},\boldsymbol{\theta'})} U(m(\underline{\theta}, \boldsymbol{\theta'})).$$

If we insert the taylor exapnsions of

- $U(\underline{\theta}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a \theta^b Q_{ab} + \mathrm{O}(\theta^3)$
- $\omega(\underline{\theta}, \boldsymbol{\theta'}) = \theta^a \theta'^b \omega_{ab} + \mathrm{O}(\theta^3)$
- $m(\underline{\theta}, \boldsymbol{\theta'}) = \underline{\theta} + \boldsymbol{\theta'} + \frac{1}{2} f^c_{ab} \theta^a \theta'^b + \mathrm{O}(\theta^3)$

we get a mess, but if we compute everything and compare the coefficients of the terms of order $\theta^a \theta'^b$ we get

$$Q_a Q_b = i m^c_{ab} Q_c + i\omega_{ab}\mathbb{I} - Q_{ab},$$

and since we wanted to compute the commutator $[Q_a, Q_b]$ we can compute

$$Q_b Q_a = i m^c_{ba} Q_c + i\omega_{ba}\mathbb{I} - Q_{ba},$$

so that subtracting the two we get

$$[Q_a, Q_b] = i(m^c_{ab} - m^c_{ba})Q_c + i(\omega_{ab} - \omega_{ba})\mathbb{I} = i f^c_{ab} Q_c + i\kappa_{ab}\mathbb{I},$$

where we have recalled the structure constants of the Lie algebra $\mathfrak{g}$ of the Lie group G and we have defined $\kappa_{ab} = \omega_{ab} - \omega_{ba}$. This last coefficients κ_{ab} are called the **central charges** of the representation U of the Lie group G. A central algebra is an algebra where the central charges are zero, so that the commutator of the generators Q_a of the representation U of the Lie group G is just

$$[Q_a, Q_b] = i f^c_{ab} Q_c.$$

The properties of the commutators of the generators Q_a of the representation U can be expressed as

- $[Q_a, Q_b] = i f^c_{ab} Q_c + i\kappa_{ab}\mathbb{I}$
- $\kappa_{ab} = \omega_{ab} - \omega_{ba}$

- $[Q_a, Q_b] + [Q_b, Q_a] = 0 \implies \kappa_{ab} + \kappa_{ba} = 0$
- $[Q_a, [Q_b, Q_c]] + [Q_b, [Q_c, Q_a]] + [Q_c, [Q_a, Q_b]] = 0$ (Jacobi identity)

The structure constants respects those identity in the following form

- $f^a_{bc} + f^a_{cb} = 0$
- $f^d_{ab} f^e_{dc} + f^d_{bc} f^e_{da} + f^d_{ca} f^e_{db} = 0$ (Jacobi identity)

and along with the central charges they respect the following identity

$$\kappa_{ad} f^d_{bc} + \kappa_{bd} f^d_{ca} + \kappa_{cd} f^d_{ab} = 0,$$

which is a consequence of the Jacobi identity for the generators Q_a of the representation U of the Lie group G. A set of coefficients f^c_{ab} and κ_{ab} which respect the above identities is called a **Lie algebra $\mathfrak{g}$ 2-cocycle**. This naming is consistent since the central charges κ_{ab} are related to the phase factor γ of the projective representation U of the Lie group G by the relation $\kappa_{ab} = \omega_{ab} - \omega_{ba}$.

...

$$U(g, g') = \gamma(g, g') U(g) U(g'),$$

$$U(g) \to \hat{U}(g) = \alpha(g) U(g),$$

$$\gamma(g, g') = \alpha(g, g') \alpha(g)^{-1} \alpha(g')^{-1},$$

so that we can redefine the unitary operators $U(g)$ of the projective representation U of the Lie group G by a suitable phase factor $\alpha(g)$ to get a new projective representation $\hat{U}$ of the Lie group G with a new phase factor $\hat{\gamma}$ which is related to the old one by the relation

$$\hat{\gamma}(g, g') = \alpha(g, g') \alpha(g)^{-1} \alpha(g')^{-1} \gamma(g, g'),$$

so that if we can find a suitable phase factor $\alpha(g)$ such that $\hat{\gamma}(g, g') = 1$ for all $g, g' \in G$, then we can redefine the unitary operators $U(g)$ of the projective representation U of the Lie group G by a suitable phase factor $\alpha(g)$ to get a new projective representation $\hat{U}$ of the Lie group G with a trivial phase factor $\hat{\gamma}$, so that the new projective representation $\hat{U}$ of the Lie group G is actually a linear representation of the Lie group G. However, in general, it is not always possible to find such a phase factor $\alpha(g)$, so that there are some projective representations of the Lie group G which cannot be redefined to get a linear representation of the Lie group G.

???

$$Q_a \to \hat{Q}_a = Q_a + k_a \mathbb{I},$$

where k_a are the central shifts of the generators Q_a of the representation U of the Lie group G. These shifts are related to the central charges κ_{ab} by the relation

$$\kappa_{ab} = f^c_{ab} k_c,$$

which defines a **2-coboundary** Lie algebra. A 2-coboundary Lie algebra is special case of a 2-cocycle Lie algebra, where the central charges κ_{ab} can be expressed as a linear combination of the structure constants f^c_{ab} with coefficients given by the central shifts k_a. This is **Lie Algebra Cohomografy**

1.4.1 | Exponential Parametrization

Starting from the exponential parametrization of the Lie group G

$$\tilde{g}(\underline{\theta}) = \lim_{n\to\infty} g(\frac{\theta}{n})^n = e^{\theta^a T_a},$$

while for the inverse and the multiplication we have

$$\begin{cases} \tilde{j}(\underline{\theta}) = -\theta^a, \\ \tilde{m}(\underline{\theta}, \underline{\theta}') = \underline{\theta} + \underline{\theta}' + \frac{1}{2} f^c_{ab}\theta^a\theta'^b + \mathrm{O}(\theta^3), \end{cases}$$

where the last function is the Baker-Campbell-Hausdorff formula (BCH) for the multiplication of the Lie group G in the exponential parametrization.

We can now use this exponential parametrization of the Lie group G to get an exponential parametrization of the projective representation U of the Lie group G:

$$\tilde{U}(g(\underline{\theta})) = U(\tilde{g}(\underline{\theta})) = U(e^{\theta^a T_a}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a\theta^b Q_{ab} + \mathrm{O}(\theta^3).$$

For the phase function ω we have

$$\tilde{\omega}(\underline{\theta}, \underline{\theta}') = \tilde{\omega}_{ab}\theta^a\theta'^b + \mathrm{O}(\theta^3).$$

At the linear level there are no differences between g and $\tilde{g}$, they only manifest from the second order in the parameters θ^a. Thus we can derive

$$Q_a = \tilde{Q}_a,$$

and for the phase function ω we have

$$\tilde{Q}_{ab} = \frac{1}{2}((\omega_{ab} + \omega_{ba})\mathbb{I} - Q_a Q_b - Q_b Q_a).$$

Now we are interested in knowing if the following is true:

$$\tilde{U}(\theta) = \exp(i\theta^a Q_a),$$

which is not the case: this is a projected representation, so this will be proven to be true only up to a phase factor. We can start from the exponential parametrization of the projective representation U of the Lie group G

$$\tilde{U}(\underline{\theta}) = U(\tilde{g}(\underline{\theta})) = \lim_{n\to\infty} U(g(\frac{\theta}{n}))^n = \lim_{n\to\infty} U((\frac{1}{n^2})^n) \neq \lim_{n\to\infty} U((\frac{1}{n^2}))^n,$$

where we highlighted the difference between this projected and a true representation. Computing further we get

$$\tilde{U}(\underline{\theta}) = \exp\left(i\sum_k^n \omega(\frac{\theta}{n}, \frac{k\theta}{n})\right) U(\frac{\theta}{n})^n,$$

where the last term, as we take the limit $n \to \infty$, becomes

$$\lim_{n\to\infty} U(\frac{\theta}{n})^n = (1 - i\frac{\theta^a}{n}Q_a)^n = e^{-i\theta^a Q_a},$$

so that the exponential parametrization of the projective representation U of the Lie group G is given by

$$\tilde{U}(\underline{\theta}) = e^{i\alpha(\underline{\theta})} e^{-i\theta^a Q_a}.$$

If we now want to get .....

$$e^{i\alpha(\underline{\theta})-i\theta^a k_a}\tilde{U}(\underline{\theta}) = e^{-i\theta^a \hat{Q}_a},$$

where $\hat{Q}_a = Q_a + k_a\mathbb{I}$ are the shifted generators of the representation U of the Lie group G: these operates in the same way since they differ only by a phase factor, but they have different central charges.

Now we found a true representation

- $\hat{\tilde{U}}(\underline{\tilde{j}}(\underline{\theta})) = \hat{\tilde{U}}(\underline{\theta})^{-1}$
- $\hat{\tilde{U}}(\underline{\tilde{m}}(\underline{\theta}, \boldsymbol{\theta'})) = \hat{\tilde{U}}(\underline{\theta})\hat{\tilde{U}}(\boldsymbol{\theta'})$

The term $\exp(-i\theta^a k_a)$ is an important **phase factor**, since it is the one which allows us to get a true representation from a projective representation by shifting the generators Q_a of the representation U of the Lie group G by a suitable central shift k_a. This is the reason why we introduced the concept of 2-coboundary Lie algebra, since it allows us to understand when a projective representation of a Lie group can be redefined to get a true representation of the Lie group.

This is a **multi-valued function** on the underlying group manifold G, since the phase factor usually is not an integer which comes back to the original value after a loop around the group manifold G (it is not guaranteed that the passage is single-valued): the exponential parametrization of the projective representation U of the Lie group G is a multi-valued function on the underlying group manifold G. Usually then we introduce a branch cut in the complex plane to make it single-valued, in order to treat multi valued function as single valued function on the more complex Riemann surface (such as for the square root and the logarithm).

$$\ln z = \ln|z| + i\arg z + 2\pi i n, \quad n \in \mathbb{Z}.$$

In our case the exponential parametrization of the projective representation U of the Lie group G is a multi-valued function on the underlying group manifold G, but can be made single-valued on a infinite Riemann surface which is a covering space of the underlying group manifold G: a covering group $\tilde{G}$. It is the same idea of the Riemann surface, but with more subtleties and complications, since the group structure has to be preserved.

There can be made analogies with the various covering groups we can find in physics, such as the universal covering group of the Lorentz group, which is the Spin group, and the universal covering group of the rotation group $SO(3)$, which is the special unitary group $SU(2)$. The projective representations of the Lorentz group and the rotation group $SO(3)$ can be redefined to get true representations of their respective covering groups, which are the Spin group and the special unitary group $SU(2)$. This is a consequence of the fact that the central charges of the projective representations of the Lorentz group and the rotation group $SO(3)$ can be expressed as linear combinations of the structure constants of their respective Lie algebras with coefficients given by suitable central shifts

$$\mathrm{SL}(2,\mathbb{C}) = \widetilde{\mathrm{SO}^+(1,3)}, \quad \mathrm{SU}(2) = \widetilde{\mathrm{SO}(3)}.$$

Example (The translation group $\mathrm{T}(3)$**).** The translation group $\mathrm{T}(3)$ implements the translation symmetry of the three-dimensional space:

$$\mathbf{a} \in \mathbb{E}_3, \quad T(\mathbf{x}) = \mathbf{x} + \mathbf{a},$$

where $T \in \mathrm{T}(3)$ and $\mathbb{E}_3$ is the three-dimensional Euclidean space.

The proper translation group T(3) can be represented by the unitary operators $U(\mathbf{a})$ for a translation $\mathbf{a} \in \mathbb{E}_3$ which act on the Hilbert space of states of a quantum system as

$$(U(\mathbf{a})\Psi)(\mathbf{x}) = \Psi(\mathbf{x} + \mathbf{a}),$$

where Ψ is a state of the quantum system and we want to prove that $\{U(\mathbf{a})\}_{\mathbf{a}\in\mathbb{E}_3}$ yields a unitary representation of the proper translation group T(3).

Proof. We can start by computing the composition of two translations **a** and **b**: [...]

There are no multiplicative phases in the right hand side. The operator family $U(\mathbf{a})$ constitutes in this way a unitary representation of the proper translation group T(3). □

Let us now discuss the infinitesimal generators of the translation group T(3). The infinitesimal generators $\hbar^{-1}\mathbf{p}$ of the unitary representation U of the proper translation group T(3) are defined through the expansion of the unitary operators $U(\mathbf{a})$ for small translations $\delta\mathbf{a}$:

$$U(\delta\mathbf{a}) = 1 + \frac{i}{\hbar}\mathbf{p} \cdot \delta\mathbf{a} + \mathrm{O}(\delta a^2),$$

with $\mathbf{p} = (p_1, p_2, p_3)$ being the momentum operator of the quantum system

$$\mathbf{p}\Psi(\mathbf{x}) = -i\hbar\nabla\Psi(\mathbf{x}).$$

Thus we can study the expansion of the unitary operators $U(\mathbf{a})$ acting on a state Ψ for small translations $\delta\mathbf{a}$:

$$\begin{aligned}(U(\delta\mathbf{a})\Psi)(\mathbf{x}) &= \Psi(\mathbf{x} + \delta\mathbf{a}) \\ &= \Psi(\mathbf{x}) + \delta\mathbf{a} \cdot \nabla\Psi(\mathbf{x}) + \mathrm{O}(\delta a^2) \\ &= \dots\end{aligned}$$

p is therefore the familiar quantum linear momentum operator as anticipated by the notation used. The result obtained is in line with the standard interpretation of momentum as the infinitesimal generator of configuration space translations.

As well–known from quantum theory, its components

$$p_k = \mathbf{p} \cdot \mathbf{e}_k, \quad k = 1, 2, 3,$$

satisfy the commutation relations

$$[p_k, p_l] = 0, \quad k, l = 1, 2, 3,$$

free of central charges, as expected from the fact that the translation group T(3) is an abelian group, so that its Lie algebra is also abelian, with vanishing structure constants $f^c_{ab} = 0$.

The translation group T(3) is a special case of a more general class of groups called the Heisenberg group, which is a non-abelian group with non-vanishing structure constants and non-vanishing central charges. The Heisenberg group is the symmetry group of the quantum phase space, and its projective representations are related to the canonical commutation relations of quantum mechanics.

Thus the operators $U(\mathbf{a})$ can be written in terms of the momentum operator **p**, the generators, as

$$U(\mathbf{a}) = e^{\frac{i}{\hbar}\mathbf{a}\cdot\mathbf{p}}.$$

Proof. limit for n.... □

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Example (The proper rotation group SO(3)/SU(2)**).**

The proper rotation group $SO(3)$ implements the rotation symmetry of the three-dimensional space:

$$R \in SO(3), \quad R : \mathbf{x} \to R\mathbf{x},$$

where R is a proper rotation and $\mathbf{x}$ is a vector in the three-dimensional space.

The proper rotation group $SO(3)$ can be represented by the unitary operators $U(R)$ for a proper rotation $R \in SO(3)$ which act on the Hilbert space of states of a quantum system as

$$(U(R)\Psi)(\mathbf{x}) = D(\mathbf{R})\Psi(R^{-1}\mathbf{x}),$$

where Ψ is a state of the quantum system, $\mathbf{x}$ is a point in the configuration space, and $D(\mathbf{R})$ is a unitary operator which acts on the internal degrees of freedom of the quantum system, such as spin.

Indeed we know that the spin of a quantum system is related to the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system; this does not come as a surprise, since the spin of a quantum system is related to the way the quantum system transforms under rotations, and the rotation group $SO(3)$ is the symmetry group of rotations in three-dimensional space.

We parametrized the representation in order to act on $\mathbf{x}$ as the inverse of the rotation,[5] since this assumption is consistent with the composition of rotations, as we can see by computing the composition of two rotations R and R':

$$U(\mathbf{R}, \mathbf{R}') = U(\mathbf{R})U(\mathbf{R}'),$$

otherwise the the order would be inverted.

$D(\mathbf{R})$ are $(2s+1) \times (2s+1)$ unitary matrices which form a representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system, where s is the spin of the quantum system

$$D(\mathbf{R})^\dagger D(\mathbf{R}) = D(\mathbf{R})D(\mathbf{R})^\dagger = \mathbb{I},$$

and they satisfy the composition law[6]

$$D(\mathbf{R}\mathbf{R}') = D(\mathbf{R})D(\mathbf{R}').$$

Proof. Let us show first that the operator $U(R)$ is unitary. Define the operator

$$U^*(\mathbf{R})\Psi(\mathbf{x}) = D(\mathbf{R})^\dagger \Psi(R\mathbf{x}),$$

so that we can compute the composition of $U^*(\mathbf{R})$ and $U(\mathbf{R})$ which acts as the identity operator on the Hilbert space of states of the quantum system:

$$(U^*(\mathbf{R})U(\mathbf{R})\Psi)(\mathbf{x}) = D(\mathbf{R})^\dagger D(\mathbf{R})\Psi(\mathbf{R}^{-1}\mathbf{R}\mathbf{x}) = \Psi(\mathbf{x}).$$

[5] Which is always well defined from the properties of groups.

[6] Here the right order is implemented by the previously discussed choice of parametrization.

The next thing to show is that

$$\|U(\mathbf{R})\Psi\|^2 = \|\Psi\|^2,$$

which is the definition of unitarity. We can compute

$$\begin{aligned}\|U(\mathbf{R})\Psi\|^2 &= \int d^3x\, |U(\mathbf{R})\Psi(\mathbf{x})|^2 = \int d^3x\, |D(\mathbf{R})\Psi(R^{-1}\mathbf{x})|^2 \\ &= \int d^3x\, |\Psi(R^{-1}\mathbf{x})|^2 = \int d^3x'\, |\Psi(\mathbf{x}')|^2 = \|\Psi\|^2,\end{aligned}$$

where we have used the unitarity of $D(\mathbf{R})$ and $J = |\det(\mathbf{R})^{-1}| = 1$. The last step is to show that the composition law is satisfied, which can be done by computing

$$\begin{aligned}(U(\mathbf{R}\mathbf{R}')\Psi)(\mathbf{x}) &= D(\mathbf{R}\mathbf{R}')\Psi((\mathbf{R}\mathbf{R}')^{-1}\mathbf{x}) = D(\mathbf{R})D(\mathbf{R}')\Psi(\mathbf{R}'^{-1}\mathbf{R}^{-1}\mathbf{x}) \\ &= D(\mathbf{R})(U(\mathbf{R}')\Psi)(\mathbf{R}^{-1}\mathbf{x}) = (U(\mathbf{R})U(\mathbf{R}')\Psi)(\mathbf{x}),\end{aligned}$$

proving that $U(\mathbf{R}\mathbf{R}') = U(\mathbf{R})U(\mathbf{R}')$. □

We now aim to study the **exponential parametrization** of the representation U of the proper rotation group $SO(3)$. The generators J_k of the representation U of the proper rotation group $SO(3)$ are defined through the expansion of the unitary operators $U(R)$ for small rotations δR.

We can then parametrize the rotation R by a vector $\boldsymbol{\varphi} = (\varphi_1, \varphi_2, \varphi_3)$ which represents the axis of rotation and the angle of rotation, so that the exponential parametrization of the the proper rotation group $SO(3)$ is given by

$$R(\boldsymbol{\varphi}) = e^{-*\varphi}, \quad R(\boldsymbol{\varphi})^{-1} = e^{*\varphi},$$

where

$$*\varphi = \begin{pmatrix} 0 & \varphi_3 & -\varphi_2 \\ -\varphi_3 & 0 & \varphi_1 \\ \varphi_2 & -\varphi_1 & 0 \end{pmatrix},$$

is the antisymmetric matrix associated to the vector $\boldsymbol{\varphi} = (\varphi_1, \varphi_2, \varphi_3)$ which parametrizes the rotation R.

Now we can reflect this exponential parametrization on the representation U of the proper rotation group $SO(3)$ to get an exponential parametrization of the representation:

$$(U(\boldsymbol{\varphi})\Psi)(\mathbf{x}) = D(\boldsymbol{\varphi})\Psi(e^{*\varphi}\mathbf{x}),$$

where $D(\boldsymbol{\varphi})$ is the exponential parametrization of the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system. We have to look at the expansion of the unitary operators $U(\boldsymbol{\varphi})$ for small rotations $\delta\boldsymbol{\varphi}$ to get the generators $\underline{j}$ of the representation U of the proper rotation group $SO(3)$:

$$U(\delta\boldsymbol{\varphi}) = 1 - i\underline{j} \cdot \delta\boldsymbol{\varphi} + \mathrm{O}(\delta\varphi^2),$$

where $\underline{j} = (J_1, J_2, J_3)$ are the generators of the representation U of the proper rotation group $SO(3)$, which are called the angular momentum operators of the quantum system. Then the expansion of the internal degrees transformation $D(\boldsymbol{\varphi})$ for small rotations $\delta\boldsymbol{\varphi}$ gives us

$$D(\delta\boldsymbol{\varphi}) = 1 - i\pm \cdot \delta\boldsymbol{\varphi} + \mathrm{O}(\delta\varphi^2),$$

where $\pm = (\Sigma_1, \Sigma_2, \Sigma_3)$ are the generators of the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system, which are called the spin operators of the quantum system. Finally we have to compute the expansion of the spatial transformation $e^{*\varphi}$ for small rotations $\delta\boldsymbol{\varphi}$ to get

$$e^{*\delta\varphi} = 1 + *\delta\varphi + \mathrm{O}(\delta\varphi^2),$$

but since the rotation can be expressed as $-\delta\boldsymbol{\varphi} \times \mathbf{x}$, we can write

$$e^{*\delta\varphi}\mathbf{x} = \mathbf{x} - \delta\boldsymbol{\varphi} \times \mathbf{x} + \mathrm{O}(\delta\varphi^2).$$

Now we can put everything together to get the expansion of the unitary operators $U(\boldsymbol{\varphi})$ for small rotations $\delta\boldsymbol{\varphi}$:

$$\begin{aligned}
(U(\delta\boldsymbol{\varphi})\Psi)(\mathbf{x}) &= D(\delta\boldsymbol{\varphi})\Psi(e^{*\delta\varphi}\mathbf{x}) \\
&= (1 - i\pm \cdot \delta\boldsymbol{\varphi} + \mathrm{O}(\delta\varphi^2))\Psi(\mathbf{x} - \delta\boldsymbol{\varphi} \times \mathbf{x} + \mathrm{O}(\delta\varphi^2)) \\
&= (1 - i\pm \cdot \delta\boldsymbol{\varphi} + i(\delta\boldsymbol{\varphi} \times \mathbf{x}) \cdot (-i\nabla) + \mathrm{O}(\delta\varphi^2))\Psi(\mathbf{x}) \\
&\ldots \\
&= \Psi(\mathbf{x}) - \delta\boldsymbol{\varphi} \cdot (\mathbf{x} \times \nabla + i\pm)\Psi(\mathbf{x}) + \mathrm{O}(\delta\varphi^2),
\end{aligned}$$

[...]

Thus we found

$$\underline{j} = \hbar\pm - i\hbar\mathbf{x} \times \nabla,$$

where renaming $\hbar\pm = \mathbf{S}$ and $-i\hbar\mathbf{x} \times \nabla = \mathbf{L}$ we get the familiar decomposition of the angular momentum operator $\underline{j}$ into the spin operator $\mathbf{S}$ and the orbital angular momentum operator $\mathbf{L}$:

$$\underline{j} = \mathbf{S} + \mathbf{L}.$$

We know that the infinitesimal generators of a continuous symmetry group satisfy the same commutation relations as the Lie algebra of the symmetry group, so that the generators J_k of the representation U of the proper rotation group $SO(3)$ satisfy the commutation relations

$$[J_k, J_l] = i\hbar\epsilon_{klm}J_m, \quad k, l, m = 1, 2, 3,$$

while the generators Σ_k of the representation of the rotation group $SO(3)$ on the internal degrees of freedom of the quantum system satisfy the same commutation relations

$$[\Sigma_k, \Sigma_l] = i\epsilon_{klm}\Sigma_m, \quad k, l, m = 1, 2, 3,$$

where ϵ_{klm} is the Levi-Civita symbol. The commutation relations of the generators J_k and Σ_k of the representation, are in line with the structure constants of the Lie algebra $\mathfrak{so}(3)$ of the proper rotation group $SO(3)$, which are given by

$$f_{kl}^{m} = \epsilon_{klm}.$$

Since we are using the exponential parametrization $\mathbf{R}(\boldsymbol{\varphi})$ of SO(3), we can get the exponential parametrization of the representation $U(\boldsymbol{\varphi})$ in terms of the infinitesimal generators J_k:

$$U(\boldsymbol{\varphi}) = e^{-\frac{i}{\hbar}\boldsymbol{\varphi}\cdot\underline{j}}.$$

Proof. limit for n.... □

It should be made an extensive digression on the universal covering group, which is the group which allows us to get a single-valued representation from a multi-valued representation by lifting the representation to the covering group. The universal covering group of a Lie group G is a simply connected Lie group $\tilde{G}$ which covers G in such a way that the projection map from $\tilde{G}$ to G is a local diffeomorphism. The universal covering group $\tilde{G}$ has the same Lie algebra as G, but it can have different global properties, such as different topology and different fundamental group. The universal covering group $\tilde{G}$ allows us to get a single-valued representation from a multi-valued representation of G by lifting the representation to $\tilde{G}$.

However we do not have the time to make this digression, so we will just mention that the universal covering group of the proper rotation group $SO(3)$ is the special unitary group $SU(2)$, which is a simply connected Lie group that covers $SO(3)$ in such a way that the projection map from $SU(2)$ to $SO(3)$ is a local diffeomorphism. The universal covering group $SU(2)$ has the same Lie algebra as $SO(3)$, but it has different global properties, such as different topology and different fundamental group. The universal covering group $SU(2)$ allows us to get a single-valued representation from a multi-valued representation of $SO(3)$ by lifting the representation to $SU(2)$.

1.5 | Dynamical Lie Symmetries and Conservation Laws

[... non lie symmetry content ...]

When a dynamical symmetry group G is a Lie group, with a parametrization $g(\underline{\theta})$, the quantum unitary representation $\tilde{U}(\underline{\theta})$ representing the action of G on the Hilbert space $\mathcal{H}$ respects

$$\tilde{U}(\underline{\theta})H\tilde{U}(\underline{\theta})^\dagger = H, \quad \forall\underline{\theta}.$$

This property restricts the form of the Hamiltonian H on one hand and has consequences for the infinitesimal generators Q_a of the unitary family $\tilde{U}(\underline{\theta})$ on the other: H commutes with the Q_a

$$[H, Q_a] = 0.$$

Proof. If we compute the derivative of the invariance condition with respect to θ^a at $\underline{\theta} = 0$, we get

$$\begin{aligned} 0 = i\frac{\partial}{\partial\theta^a}(H - \tilde{U}(\underline{\theta})H\tilde{U}(\underline{\theta})^\dagger)\Big|_{\underline{\theta}=\underline{0}} &= \left[-i\frac{\partial\tilde{U}(\underline{\theta})}{\partial\theta^a}H\tilde{U}(\underline{\theta})^\dagger + \tilde{U}(\underline{\theta})H\left(i\frac{\partial\tilde{U}(\underline{\theta})}{\partial\theta^a}\right)^\dagger\right]_{\underline{\theta}=\underline{0}} \\ = -i\frac{\partial\tilde{U}(\underline{0})}{\partial\theta^a}H\tilde{U}(\underline{0})^\dagger + \tilde{U}(\underline{0})H\left(i\frac{\partial\tilde{U}(\underline{0})}{\partial\theta^a}\right)^\dagger &= -Q_aH + HQ_a = [H, Q_a], \end{aligned}$$

since $\tilde{U}(\underline{0}) = \mathbb{I}$ and $Q_a = -i\frac{\partial\tilde{U}(\underline{\theta})}{\partial\theta^a}\big|_{\underline{\theta}=\underline{0}}$. This proves that when a system is invariant under a Lie group of dynamical symmetries, the generators of that group commute with the Hamiltonian. □

This result is significant because it implies that the generators Q_a are conserved quantities of the system, as they commute with the Hamiltonian. In quantum mechanics, this means that the eigenvalues of Q_a are **constants of motion**, and the system's dynamics can be analyzed in terms of these conserved quantities. This connection between symmetries and conservation laws is a fundamental aspect of both classical and quantum physics, encapsulated in **Noether's theorem**. Indeed, we can defin the evolution operator

$$U(t,s) = e^{-\frac{i}{\hbar}(t-s)H},$$

then we have

$$U(t,s)Q_aU(t,s)^\dagger = e^{-\frac{i}{\hbar}(t-s)H}Q_ae^{\frac{i}{\hbar}(t-s)H} = Q_a,$$

which shows that Q_a is conserved under the time evolution generated by H.

Proof. This is a direct consequence of the commutation relation $[H, Q_a] = 0$... □

We have just shown the quantum analog of Noether's theorem: for every continuous symmetry of the action (or equivalently, for every Lie group of dynamical symmetries), there is a corresponding conserved quantity. This profound connection between symmetries and conservation laws is a cornerstone of modern physics, providing deep insights into the fundamental structure of physical theories.

Appendices

A | Notation and Conventions

B | Computations and further developments